$$u_{\mathcal{P}}(\theta) = [u_\theta(r_1), \ldots, u_\theta(r_{175})], \quad p = \Psi(u_{\mathcal{P}})$$

$$\nabla_\theta L_B = J_{u_{\mathcal{P}},\theta}^{\top} J_{\Psi}^{\top} J_H^{\top} \nabla_{\widehat{B}} L_B$$